GHOST PROTOCOL — MASTER IMPLEMENTATION PLAN

Comprehensive A–Z System Architecture, Codebase Reuse Map, Subsystem Blueprint, Data Schemas, 14-Phase Implementation Order, and 20-Scenario Docker Lab Test Suite

> **Master Mandate for Antigravity:** This specification governs the complete engineering implementation of **GHOST PROTOCOL**—an autonomous cyber-threat observation, search, correlation, and explanation platform. GHOST Protocol builds an intelligent, graph-connected memory of threat actors, aliases, ransomware campaigns, data breach claims, malware, and infrastructure changes over time. **Constraint:** All existing DarkWatch machine learning models (DistilBERT, XGBoost, Temporal Autoencoder v2, HM-TIF Fusion), CTI/STIX engines, MITRE mappers, and Whonix isolation boundaries remain 100% frozen with zero retraining or architectural redesign.

1. GHOST as an Autonomous AI Agent

GHOST operates as an autonomous orchestration and reasoning layer over a closed loop: **DISCOVER** → **VALIDATE** → **OBSERVE** → **UNDERSTAND** → **EXTRACT** → **CORRELATE** → **REMEMBER** → **INVESTIGATE** → **ANALYZE** → **EXPLAIN** → **ALERT**.

Unlike simple web scrapers or static CVE matchers, GHOST maintains state across operational sessions, handles session-gated access barriers via authorized browser sessions, resolves entity aliases across changing infrastructure, and correlates evidence across multiple public platforms (Telegram, darknet forums, leak portals, paste sites, and search indexes).

> **Agent Execution Loop:**
> 1. **Discovery & Validation:** Monitors seed feeds (Daunt.link, deepdarkCTI, Ahmia) and validates reachability via Whonix SOCKS5 (socks5h://10.152.152.10:9050).
> 2. **Observation & Preprocessing:** Fetches static HTML (JS-disabled default) or executes authorized 2-step Playwright sessions (Dread). Cleans text with Trafilatura, masks PII/Crypto/Onion tokens, and computes SHA-256 content hashes and MinHash LSH signatures.
> 3. **Multi-Modal ML Inference:** Routes evidence to frozen DarkWatch ML models (DistilBERT, XGBoost, Temporal AE v2) and HM-TIF Cross-Attention Fusion to compute Dynamic Risk Scores (0.00–1.00).
> 4. **Graph Memory & Entity Resolution:** Extracts CTI entities (TIJERE), executes Composite Similarity Matching (Jaro-Winkler + Cosine Vector + Graph Jaccard), tracks infrastructure evolution, and updates the Temporal Property Graph in SQLite (darkwatch.db).
> 5. **Investigation & Explanation:** Supports natural-language analyst console queries, triggers Tier 1 token highlights and Tier 2 SHAP charts (Risk ≥ 0.70), and executes transient Sherlock OSINT handle checks.

2. Codebase Audit & Component Reuse Map

Antigravity must build directly upon the existing repository structure without creating parallel architectures or duplicate logic.

Category	File / Module	Status	Implementation Responsibility
ML Models	`darkwatch/models/text_inference.py`	REUSE (Frozen)	HackerSignal DistilBERT (8 exploit classes, 91.3% acc, 768-dim vector)
ML Models	`darkwatch/models/network_inference.py`	REUSE (Frozen)	Multiclass XGBoost (3 transport classes, 64-dim feature vector)
ML Models	`darkwatch/models/temporal_inference.py`	REUSE (Frozen)	Temporal Autoencoder v2 (14 inputs -> 8-dim latent z, MSE loss)
Fusion Engine	`darkwatch/fusion/fusion_engine.py`	REUSE (Frozen)	HM-TIF Hierarchical Cross-Attention with missing-modality resilience
CTI & STIX	`darkwatch/cti/stix_generator.py`	REUSE (Frozen)	OASIS STIX 2.1 JSON Bundle Generator (SDOs, SCOs, SROs)
MITRE ATT&CK;	`darkwatch/cti/mitre_mapper.py`	REUSE (Frozen)	Enterprise ATT&CK; T-Code Resolver (T1190, T1486, mitigations)
Explainable AI	`darkwatch/xai/xai_generator.py`	REUSE (Frozen)	Tier 1 Token Highlights + Tier 2 SHAP Bar Charts (Risk ≥ 0.70)
Preprocessing	`darkwatch/sanitizer.py`	REUSE / MODIFY	Trafilatura cleaning, token masking, SHA-256 & MinHash LSH deduplication
Pipeline Bridge	`darkwatch/pipeline/lab_bridge.py`	MODIFY	In-memory ingestion bridge connecting Stage 1 outputs to DarkWatch ML
Storage / DB	`darkwatch/storage/annotation_store.py`	MODIFY	SQLite GoldRecord store; extend schema for graph, entities, sources
Source Registry	`darkwatch/discovery/source_registry.py`	NEW	Source lifecycle database manager and discovery queue
Discovery	`darkwatch/discovery/search_discoverer.py`	NEW	Ahmia search indexer, deepdarkCTI feed parser, Daunt.link mirror resolver
Telegram Observer	`darkwatch/observer/telegram_public_observer.py`	NEW	Public HTML web preview parser for t.me/s/SliceForLifeee (No login)
Dread Observer	`darkwatch/observer/dread_session_observer.py`	NEW	Playwright SOCKS5 session manager for Dread 2-step & cookie vault
Link Extractor	`darkwatch/discovery/link_extractor.py`	NEW	Recursive regex link extractor for outbound .onion and t.me URLs
Entity Resolution	`darkwatch/resolution/entity_resolver.py`	NEW	Composite matching engine for actor handles, aliases, and infra changes
Sherlock OSINT	`darkwatch/investigation/sherlock_wrapper.py`	NEW	Transient surface-web handle lookup wrapper (Omitted from DB)
Console & API	`darkwatch/dashboard/app.py`	NEW	SOC Analyst Workstation, Natural Language Query Console, Alert Cards

3. Human-Language Understanding & Threat Semantics

GHOST must understand threat intent even when the author omits technical jargon or explicit keywords like 'CVE' or 'exploit'. The system processes natural-language threat semantics across six core threat domains:

Threat Category	Natural Language Indicators	Extracted CTI Entities	STIX 2.1 Mapping
Data Breach Claims	database dump, 50GB customer records, employee credentials, stolen sql	Victim Org, Data Volume, Schema, File Hashes	observed-data, indicator, report
Access Brokerage	admin panel access, rdp credentials, domain admin shell, corporate vpn	Target Industry, Access Level, Price Quotes	infrastructure, tool, threat-actor
Ransomware Notices	network encrypted, lockbit 3.0 leak, pay bitcoin, victim publication	Victim Org, Ransom Deadline, Crypto Wallets	malware, threat-actor, course-of-action
RCE & Exploits	zero-day pre-auth rce, proof of concept, bypass authentication, unpatched	Target System, Affected Versions, CVE ID	vulnerability, attack-pattern, tool
Credential Dumps	stealer logs, redline logs, combo list, login list, user/pass dump	Exposed Usernames, Domains, File Signatures	indicator, identity, file
Malware Distribution	crypter stub, silent stealer, fud loader, reverse shell payload	Malware Family, File Hashes, C2 Domains	malware, infrastructure, indicator

4. Subsystem Specifications

4.1 Whonix Isolation & Security Boundary

All real-world observation traffic must pass through the **Whonix-Gateway SOCKS5 proxy (socks5h://10.152.152.10:9050)**. Direct clearnet socket creation is strictly forbidden. Before every outbound fetch, `whonix_collector.py` executes a pre-flight socket test (_verify_whonix_gateway()). If the proxy is unreachable, execution halts immediately with a WhonixIsolationError, enforcing a fail-closed network boundary.

4.2 Hybrid Dual-Engine Observation Architecture

Observation separates bulk polling from interactive session bootstrap:
• **Engine A (JS-Disabled Bulk Polling):** Uses httpx.AsyncClient over SOCKS5 with JavaScript disabled by default (`Accept: text/html`). Minimizes Tor latency and eliminates client-side script execution risks.
• **Engine B (Playwright Session Observer):** Used exclusively for session-gated platforms like **Dread**. Resolves Dread's live mirror via Daunt.link, completes the authorized 2-step verification flow, and exports valid session cookies into an **In-Memory Volatile Vault**. Raw secrets are omitted from disk; only SHA-256 cookie hashes are written to `darkwatch.db`.

4.3 Entity Resolution & Infrastructure Tracking

When an actor handle or entity appears across observations, `entity_resolver.py` calculates a Composite Similarity Score:
S_comp = 0.40 * JaroWinkler(E_new, E_canon) + 0.35 * CosineSim(V_new, V_canon) + 0.25 * GraphJaccard(N_new, N_canon)
• **S_comp ≥ 0.85:** Automatic Merge — Merges entity into canonical record, updates `last_seen_utc`, and increments co-occurrence.
• **0.65 ≤ S_comp < 0.85:** Candidate Alias Link — Instantiates candidate alias in `entity_aliases` with supporting evidence.
• **Infrastructure Evolution Example:** When *ShadowFox* moves from abc.onion (offline) to xyz.onion (active), the

engine links the new infrastructure, creates a REPLACED_BY edge, and updates provenance.

4.4 On-Demand Sherlock OSINT Engine

When an analyst requests a handle search, `sherlock_wrapper.py` queries surface-web platforms (GitHub, forums, keybases) asynchronously. Profile matches are rendered live on the dashboard but **omitted from permanent `darkwatch.db` storage**, maintaining dataset compliance and preventing database clutter.

5. Extended SQLite Database Schema (darkwatch.db)

`annotation_store.py` manages the following extended tables in `darkwatch.db`:

Table Name	Primary Keys	Key Fields & Data Types	Description
sources_registry	source_id	canonical_url (TEXT), validation_status (TEXT), last_checked_utc (TIMESTAMP), raw_html_sha256 (TEXT), clean_text_sha256 (TEXT), minhash_sig (BLOB)	Tracks target sources, reachability, content hashes, and LSH signatures.
entities	entity_id	canonical_name (TEXT), entity_type (TEXT), first_seen_utc (TIMESTAMP), last_seen_utc (TIMESTAMP), observation_count (INT), text_vector (BLOB)	Stores canonical threat entities (actors, companies, malware, infrastructure).
entity_aliases	alias_id	canonical_entity_id (FK), alias_name (TEXT), similarity_score (REAL), evidence_text (TEXT), status (TEXT: CANDIDATE/CONFIRMED)	Records actor aliases, monikers, and candidate identity matches.
graph_edges	edge_id	source_entity_id (FK), target_entity_id (FK), relation_type (TEXT), confidence (REAL), provenance_obs_id (FK)	Stores directed property graph edges (OPERATES, USES, TARGETS, REPLACED_BY).
breach_claims	claim_id	victim_organization (TEXT), threat_actor (TEXT), data_volume (TEXT), crypto_wallet (TEXT), claim_timestamp_utc (TIMESTAMP)	Records data breach and ransomware extortion claims without payload dumps.

6. 14-Phase Implementation Order for Antigravity

Phase	Phase Name	Core Implementation Deliverables
Phase 1	Repository Audit	Verify existing ML imports (DistilBERT, XGBoost, Temporal AE, HM-TIF) and test suite.
Phase 2	Database Schema	Extend darkwatch.db schema in annotation_store.py (sources, entities, graph_edges).
Phase 3	Whonix Boundary	Implement socket test in whonix_collector.py; enforce fail-closed socks5h proxy.
Phase 4	Source Registry	Implement source_registry.py manager; ingest Ahmia, deepdarkCTI, Daunt.link.
Phase 5	Telegram Observer	Implement telegram_public_observer.py for t.me/s/SliceForLifeee HTML web preview.

Phase	Phase Name	Core Implementation Deliverables
Phase 6	Dread Session Observer	Implement dread_session_observer.py; Playwright SOCKS5, 2-step verification, cookie vault.
Phase 7	Link Extractor	Implement link_extractor.py regex URL parser and pending discovery queue.
Phase 8	Threat Filter	Implement threat_filter.py natural-language pre-filter and regex CTI tagger.
Phase 9	Pipeline Bridge	Modify lab_bridge.py to accept Stage 1 outputs and run frozen DarkWatch ML.
Phase 10	Entity Resolver	Implement entity_resolver.py composite matching score and alias correlation.
Phase 11	Sherlock OSINT	Implement sherlock_wrapper.py transient surface-web handle lookup.
Phase 12	Graph Correlation	Implement CTI graph traversal queries and STIX 2.1 relationship builders.
Phase 13	Analyst Dashboard	Implement app.py Flask dashboard, Explained Alert Cards, and NL Query Console.
Phase 14	Docker Lab Demos	Execute 20-Scenario Docker lab test suite and verify end-to-end acceptance criteria.

7. 20-Scenario Docker Lab Test Suite

Antigravity must verify the system against these 20 explicit testing scenarios using `/workspace/darkwatch_labs/`:

#	Scenario Name	Lab Endpoint / Target	Expected System Output & Acceptance Criteria
1	Public Forum Crawl	/forum	HTTP 200; clean text extracted; DistilBERT threat category returned.
2	Marketplace Access	/market	Extracted price quotes, vendor handles, and masked [ID_CRYPTO] addresses.
3	Paste Indicator Dump	/paste	Extracted SHA-256 hashes, CVE mentions, and raw code snippets.
4	Ransomware Extortion	/leak	Extracted victim org, 50GB claim, LockBit tag, and STIX 2.1 bundle.
5	503 Barrier Handling	/barrier	AUTOMATED_ACCESS_BLOCKED detected; HM-TIF maintains risk score via net/temporal.
6	Dread 2-Step Session	/session_gated	Playwright completes 2-step; cookie exported to vault; status SESSION_ACTIVE.
7	Dynamic Content Delta	/dynamic_forum	Content diff detected; clean_text_sha256 updated; new post ingested.
8	Near-Duplicate Mirror	/forum_mirror	MinHash LSH $J \geq 0.80$; flagged NEAR_DUPLICATE_MIRROR and skipped.
9	Recurring Threat Actor	Lab Post with 'ShadowFox'	Entity match $S_{comp} \geq 0.85$; observation_count incremented; last_seen updated.
10	Infrastructure Shift	ShadowFox on xyz.onion	REPLACED_BY edge created between abc.onion and xyz.onion; provenance logged.

#	Scenario Name	Lab Endpoint / Target	Expected System Output & Acceptance Criteria
11	Telegram Web Preview	t.me/s/SliceForLifeee	No-login HTML fetch; parsed post IDs, timestamps, and embedded onion links.
12	Daunt Mirror Resolve	daunt.link/dread	Resolved active signed onion address for Dread and updated registry.
13	deepdarkCTI Feed Ingest	fastfire GitHub Feed	Ingested ransomware_gang.md; queued new onion URLs for validation.
14	Sherlock Handle OSINT	Analyst query 'R3dDevil'	Transient profile matches returned to UI; omitted from permanent DB.
15	DistilBERT Classifier	Exploit text input	Returned top class (e.g., 'rce', 91.3% acc) and 768-dim latent vector.
16	XGBoost Classifier	Network PCAP input	Returned transport class ('Tor', 'VPN', 'Clearnet') and 64-dim vector.
17	Temporal Autoencoder	14 session features	Returned reconstruction MSE loss, anomaly flag (threshold_95), 8-dim z.
18	Two-Tier XAI Display	Risk Score ≥ 0.70	Tier 1 token highlights + Tier 2 SHAP feature attribution bar charts rendered.
19	NL Console Query	'Show LockBit victims'	Returned graph evidence, victim orgs, wallets, and STIX export button.
20	Whonix Isolation Fail	SOCKS5 Proxy Offline	Pre-flight socket check fails; WhonixIsolationError raised; zero IP leakage.

8. Module Specification Blueprints

> **Standardized Engineering Specification Format for Antigravity:**
> Every module created or modified must implement the following attributes:
> • **PURPOSE:** Core operational responsibility.
> • **INPUT:** Explicit parameters, data types, and schemas.
> • **PROCESS:** Algorithm, logic flow, and error handling.
> • **OUTPUT:** Data structures returned or published.
> • **STORAGE:** Tables written to in darkwatch.db or volatile memory.
> • **DEPENDENCIES:** Required Python packages or internal modules.
> • **SECURITY:** Boundary enforcement, sanitization, and secret hygiene.
> • **TEST:** Corresponding unit test script in darkwatch/tests/.